Kim Knudsen, Technical University of Denmark

Inverse Problems and Imaging (02624)

Week 7

Bayes' formula

Mathematical description requires probability space (Ω, σ, P).

Joint probability, conditional and marginal of events $(E_1, E_2 \in \sigma)$

$$P(E_1 \cap E_2) = P(E_1|E_2)P(E_2).$$

Stochastic variables $X : \Omega \to \mathbb{R}^n$ with measure (of Borel set B)

$$\mu(B) = P(X^{-1}(B)).$$

For two stochastic variables we have joint, conditional and marginal measures

$$\mu(B_1, B_2) = \mu(B_1|B_2)\mu(B_2)$$

leading to Bayes' formula

$$\mu(B_1|B_2) = \frac{\mu(B_2|B_1)\mu(B_1)}{\mu(B_2)}.$$

Bayesian Inverse Problems

At the level of probability densities π we can make sense of a similar formula

$$\pi(x_1|x_2) = \frac{\pi(x_2|x_1)\pi(x_1)}{\pi(x_2)}.$$

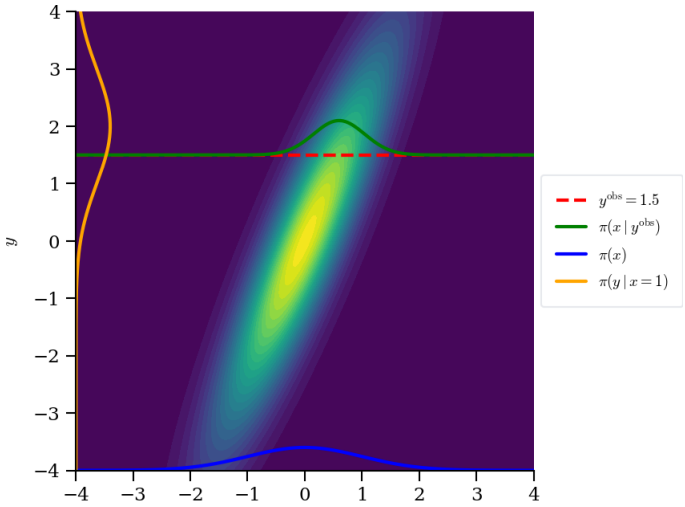
Put $x_1 = x$ and $x_2 = y^\delta$. We cast the problem $Ax = y^\delta$ for stochastic variables X with prior $\pi_{\text{pr}}(x)$, observation y_{obs} with likelihood $\pi(y_{\text{obs}}|x)$, and interpret Bayes' formula for the posterior

$$\pi_{\text{post}}(x) = \pi(x|y_{\text{obs}}) = \frac{\pi(y_{\text{obs}}|x)\pi_{\text{pr}}(x)}{\pi(y)}.$$

Solution to the inverse problems is a probability density of (possible) solutions under the prior and likelihood.

Illustration: Bayesian inverse problem

Model $y^\delta = 2x + e$, with Gaussian prior and likelihood



Exploring the posterior distribution

Estimators x_{MAP} and x_{CM}. For the posterior based on bimodal Gaussian

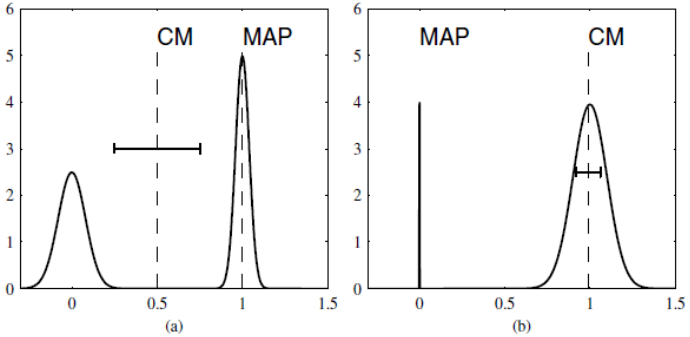


Figure 3.1. Examples of x_{CM} and x_{MAP}. The figures represent the probability densities $\pi(x)$ with parameter values $\alpha = 1/2$, $\sigma_0 = 0.08$ and $\sigma_1 = 0.04$ (a), and $\alpha = 0.01$, $\sigma_0 = 0.001 = \alpha\sigma_1$ and $\sigma_1 = 0.1$ (b).

Kaipio and Somersalo, 2005.

Gaussian case

Assuming $Ax = y + e$ with prior and noise model

$$X \sim \mathcal{N}(0, \gamma^2 I),$$
$$E \sim \mathcal{N}(0, \sigma^2 I).$$

Bayes' formula then shows that the posterior is Gaussian with density

$$\pi_{\text{post}}(x) = \pi(x|y_{\text{obs}}) = \frac{\pi(y_{\text{obs}}|x)\pi_{\text{pr}}(x)}{\pi(y)} \propto \exp\left(-\frac{1}{2}(x-\overline{x})^T \Gamma_{\text{post}}^{-1}(x-\overline{x})\right).$$

Here

$$\Gamma_{\text{post}} = \gamma^2 - \gamma^4 A^T(\gamma^2 AA^T + \sigma^2)^{-1}A,$$
$$\overline{x} = R_\alpha y_{\text{obs}},\ \alpha = \frac{\sigma^2}{\gamma^2}.$$

More to say?

- Rigourous treatment requires measure theory
- Bayesian Inverse Problems are most often well-posed
- Rich modelling framework
- No assumption regarding linearity
- Exploring the posterior requires sampling strategies - CUQI
- Computationally hard for large scale problems